

Assignment - 3

Aim: Implement optimization algorithms.

Theory

1) Momentum-Based GD

⇒ Enhances standard gradient descent by adding a fraction of the previous descent by adding a update to the current one, helping to accelerate convergence and smooth out oscillations.

2) Nesterov Accelerated GD

⇒ An improved version of momentum that calculates the gradient at the anticipated next position, resulting in more accurate and responsive updates.

3) Adagrad

⇒ Adapts the learning rate for each parameter individually based on the historical sum of squared gradients, allowing larger updates for infrequent parameters.

4) RMSProp

⇒ Modifies Adagrad by using an exponentially decaying average of squared gradients to avoid the rapid decrease in learning rate, ensuring better performance on non-stationary problems.

5) Adam

⇒ Combines the benefits of Momentum and RMSProp by using both the moving average of gradients and the squared gradients to adapt learning rates for each parameter.



~~Adam~~